Ishaan Desai

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Education

Northeastern University — Boston, MA
Master of Science in Electrical and Computer Engineering — *Expected May 2027*
- **GPA: 4.0** — Concentration: Machine Learning, Computer Vision, and Algorithms
- Coursework: Large Language Models, Fundamentals of Computer Engineering, High Performance Computing, Computer Vision, Machine Learning, Robotics Sensing & Navigation

Northeastern University — Boston, MA
Bachelor of Science in Computer Engineering, Minor in Business Admin. — *Dec. 2025*
- GPA: 3.45

Technical Experience

CardioFocus LLC. — Jan 2025 – June 2025
R&D Software & Systems Engineer Co-op — *Marlborough, MA*
- Designed and implemented a moving average algorithm in **C#** for the QuickShot PFA system's contact detection app to improve real-time tissue contact sensing during atrial fibrillation procedures.
- Built **Python (Pandas, Plotly)** telemetry analysis pipeline processing generator logs to validate power board design decisions with statistical evidence.
- Designed system architecture and UI/UX workflows for next-generation control interface, translating clinical requirements into engineering specifications.

Insulet Corporation — Jan 2024 – Aug 2024
Systems Engineer, Design Verification Co-op — *Acton, MA*
- Collaborated with **Cloud/SRE** team to validate API reliability and system integration for Omnipod 5 cloud infrastructure ahead of a **France market launch**.
- Developed **VBA** automation tool for lab inventory management, reducing manual technician workload by **80%** and streamlining equipment tracking.
- Executed end-to-end validation of iOS mobile application and onboarded incoming co-ops, ensuring reliability standards were met prior to global release.

TransMedics Inc. — Jan 2023 – Aug 2023
Software Quality Assurance Engineer Co-op — *Andover, MA*
- Enhanced **C++** embedded codebase by fixing memory logic bugs, standardizing sensor measurement classes, and updating alarm systems to improve Organ Care System stability.
- Automated data extraction workflows using **Python**, parsing binary sensor logs and refactoring **1,000+ source file** headers, reducing manual processing time by **90%**.
- Authored version control migration protocol (Bitbucket Server to Data Center) ensuring zero data loss for distributed engineering team.

Projects

SkySense: Multi-Sensor Drone Detection System (3rd/19 Teams) — July 2025 – Dec 2025
- Trained a custom **YOLOv8 Nano** model on a **20,000+** image dataset using advanced augmentation, achieving **99.5% mAP** and **100% recall** for autonomous threat detection.
- Deployed the model on an **NVIDIA Jetson Orin Nano** using **TensorRT** quantization to process live video at **21 FPS**, enabling real-time edge computing performance.
- Led model training, deployment, and web infrastructure as **ML/Web lead** on a 6-person team, architecting a multi-threaded Flask server ingesting async ROS2 topics and rendering live MJPEG streams at 5 Hz.

Parallel Monte Carlo European Options Pricing — Jan 2026 – Apr 2026
- Engineered a Monte Carlo options pricer in **Serial C++, OpenMP, and CUDA**, benchmarking **10K–100M** simulated paths on an **NVIDIA Tesla P100** cluster, validating pricing accuracy against Black-Scholes to within **$0.01**.
- Achieved a **6.4x speedup** via OpenMP and implemented **antithetic variates** variance reduction, cutting RNG-phase runtime **3x** by targeting a cuRAND initialization bottleneck identified through phase-level profiling.

Panoramic Scene Reconstruction with Object Detection & Tracking — Jan 2026 – Apr 2026
- Built a 3-stage panoramic video stitching pipeline (**SIFT**, **RANSAC** homography estimation, multi-band Laplacian pyramid blending) achieving complete 29/29-pair homography chains across two SportsMOT clips.
- Integrated **YOLOv8n-seg** detection with cross-frame **NMS** and **ByteTrack** multi-object tracking, merging **15–35%** duplicate detections and producing color-coded activity maps validated with SSIM overlap metrics.

Technical Skills

Languages: C++, Python, C#, SQL, MATLAB, JavaScript, HTML/CSS
Parallel/HPC: CUDA, OpenMP, MPI, SLURM
Frameworks: ROS2, PyTorch, Ultralytics YOLO, TensorRT, Flask, React
Developer Tools: Git, Linux/Unix, Docker, VS Code, Jira, Postman
Libraries: Pandas, Plotly, OpenCV, SciPy, STL, NumPy